

Statistics for Data Analysis

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Agenda

- Probability
- Bayes Theorem
- Application to ML

Statistics....

- Statistics is NOT only numbers and figures. The numbers may be right, but the interpretation may be wrong.
- Statistics refer to a range of techniques and procedures for collecting, analysing, interpreting, displaying and making decision based on data.

“Every time I buy a vanilla ice-cream, when I start back from the store my car won't start”

[Source: Vanilla Ice Cream caused General Motors to not start | LinkedIn](#)

Statistics....

- Deterministic event: This is when an experiment could tell us precisely what will happened under certain conditions. For instance, if we put in a jar with one litre of water in it on the fire at 100 degrees Celsius, we know that it will boil after 3 minutes, and if we repeat this experiment many times (**under the same conditions**), we will always get the same result (or small variations).
- Stochastics (Probabilistic) event: This is when an experiment depend on chance, and therefore we could get always different results even though we repeat it under the same conditions. Level of Randomness.

Statistics is part of our lives....

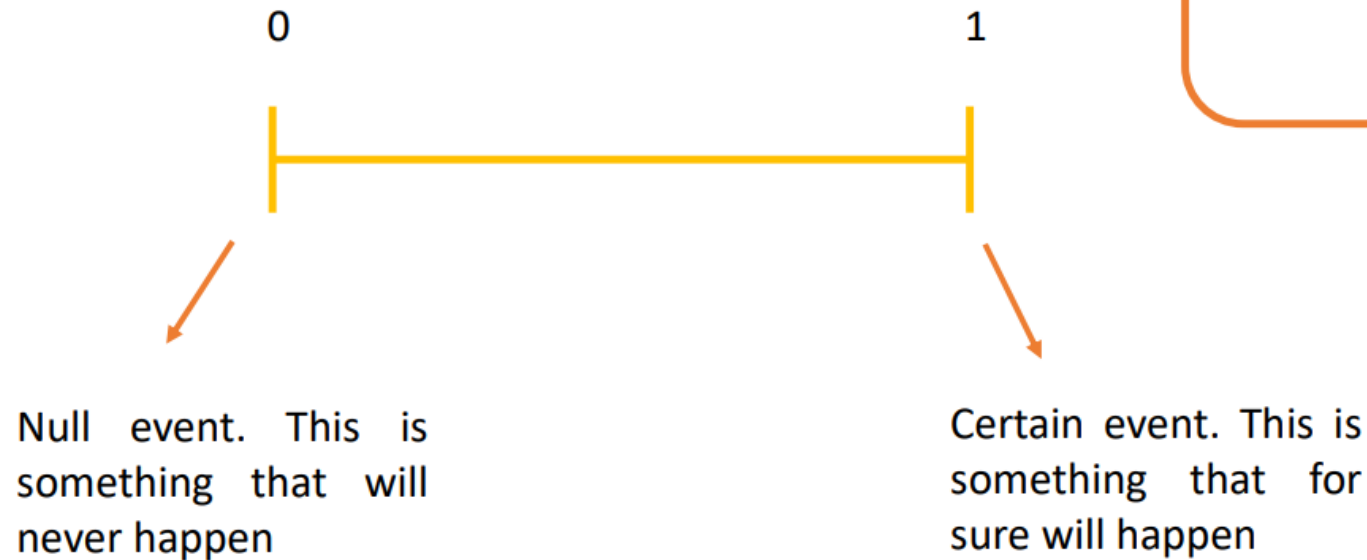
We find Statistics every day around us and maybe we do not realise. We are talking about Statistical Events when:

- We check the weather on our phone and it says that today there is a 70% chances of rain/have a sunny day.
- We mention that 9% of the global population live in poverty.
- We play the lottery.
- Choose a card from a deck.
- Discuss about possible results in sports.
- When we discuss about the prices.



1. Probabilities

Probabilities can take values between 0 and 1.



A probability could be zero and could be one



1. Probabilities

The events are named with a capital letter: A, B, C, D...

For example:

A = picking a white ball from a bag

B = pass an exam

And so on.

Statistic is a linguistic problem! 

Colloquial Language	Symbol	Probability	Operation
or	$\cup$	Union	Addition (+)
and	$\cap$	Intersection	Multiplication (*)

*We multiply only
when the events
are independent.*

1. Probabilities

Independent Events

The outcome of one event **does not** affect the outcome of the other.

If A and B are independent events then the probability of both occurring is

$$P(A \text{ and } B) = P(A) \times P(B)$$

Dependent Events

The outcome of one event affects the outcome of the other.

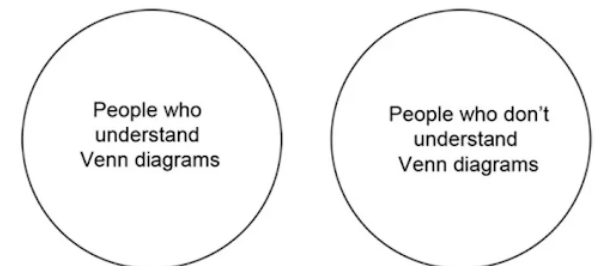
If A and B are dependent events then the probability of both occurring is

$$P(A \text{ and } B) = P(A) \times P(B | A)$$

Probability of B given A

Mutual exclusive events: These are those events that cannot happen at the same time.

Venn-Venn



1. Probabilities

This is the basic formula to calculate a probability:

$$P(A) = \frac{\text{Favorable cases of the event A}}{\text{Total cases}}$$

Suppose that in this classroom we have 30 males and 20 females, and we want to randomly choose a person, what is the probability of choosing a male?

A = the person is a male

$$P(A) = \frac{30}{50} = 0.60$$

Number of males in the group

Total people in the group

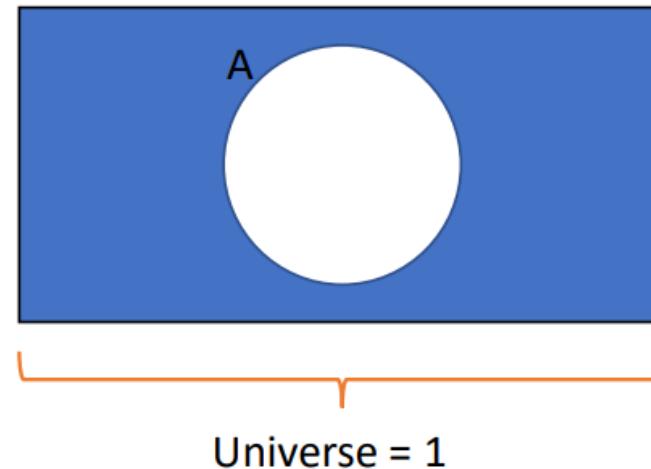
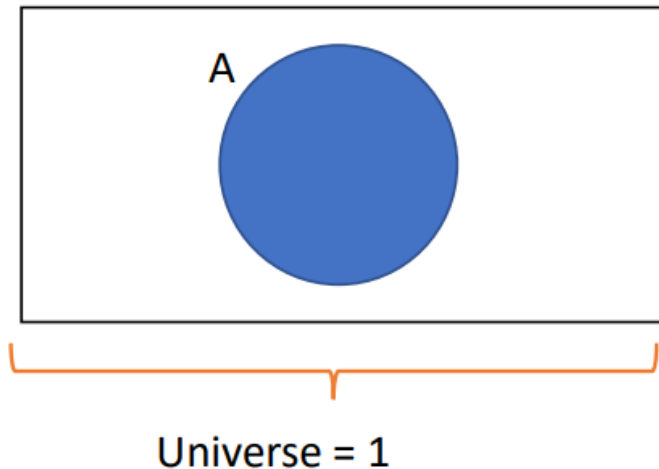
60% of the students are males.

1. Probabilities

These axioms are auxiliars that help us to solve problems related to probabilities. There are 3 basic Axioms that we will commonly use.

1) Complementary probability.

If we know the probability of A , we could know the probability of $\underline{A}$



1. Probabilities

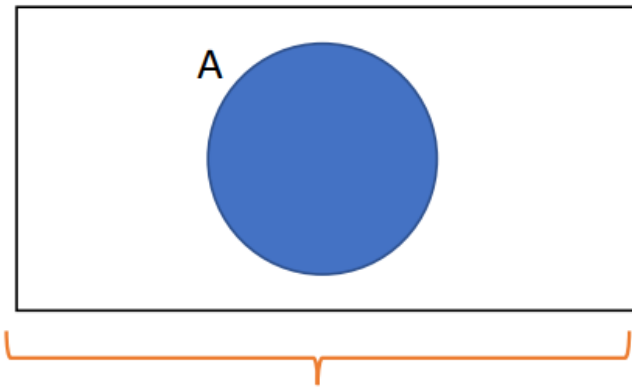
1) Complementary probability:

If the probability of my universe is 1 and I know the probability of A, I can find the probability of A

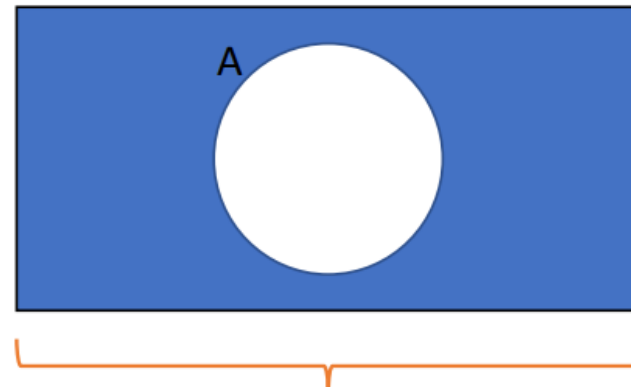
$$P(\underline{A}) = 1 - P(A)$$

And the other way around

$$P(A) = 1 - P(\underline{A})$$



Universe = 1

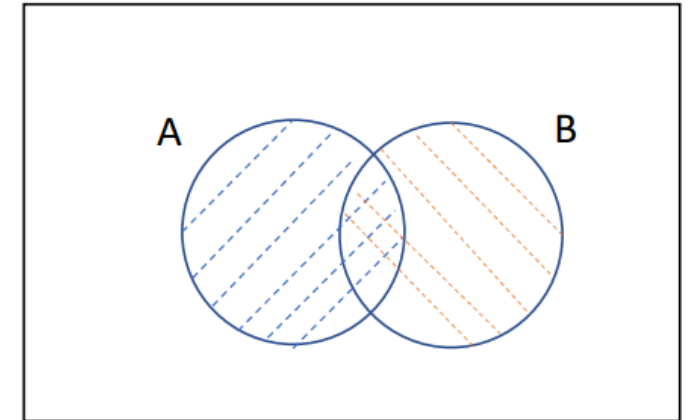
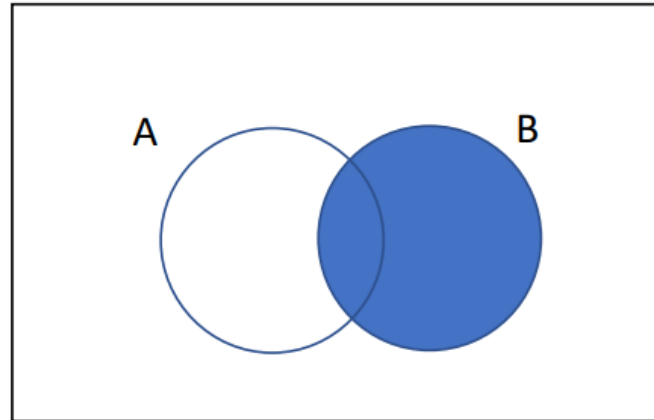
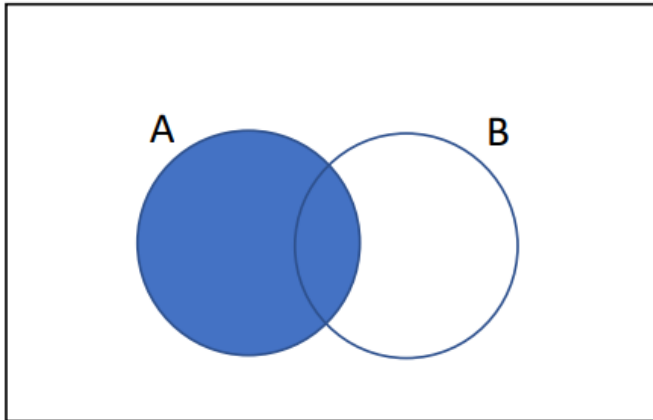


Universe = 1

1. Probabilities

2) Axiom of the addition:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



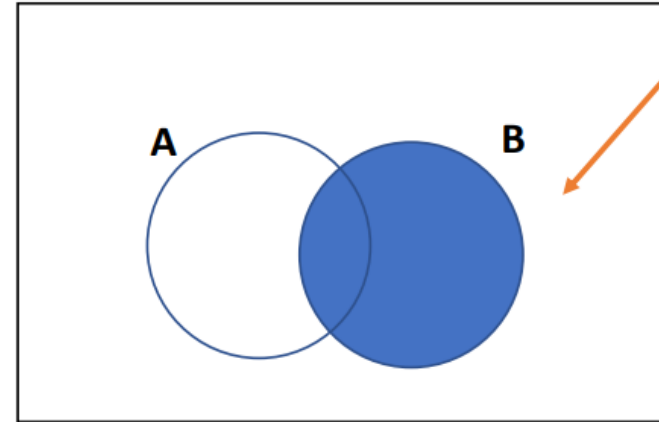
1. Probabilities

3) Conditional probability:

If we know something about the data, we do not need to use the full group to find our solution.

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

The slash means knowing that...
Given that...
Assuming that...



If we know that the observation belongs to B, we do not need to take into consideration what happens with A.

1. Probabilities

Contingency Tables: We use them when there is an idea of **simultaneity** in our data.

You are the owner of a start-up company, and you need to hire people. In order to do so, you post an advertising on a social media and 100 candidates send their CV: 30 of them do not have a third level degree, while the rest do, and 60 candidates speak a foreign language. Also, 10 of the candidates do not have a third level degree but they speak a foreign language. If you randomly choose a candidate:

- a) What is the probability of choosing a person who has a third level degree?
- b) What is the probability of choosing a person who does not speak a foreign language?
- c) What is the probability of choosing a person who has a third level degree or speak a foreign language?
- d) If we know that the person speaks a foreign language, what is the probability that also holds a third level degree?

1. Probabilities

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D = the person holds a third level degree

F = the person speaks a foreign language

	D	<u>D</u>	TOTAL
F		10	60
<u>F</u>			
TOTAL		30	100

We need to check the information available and build the table. Don't forget to define your events



1. Probabilities

Contingency Tables: We use them when there is an idea of simultaneity in our data.

c) What is the probability of choosing a person who has a third level degree or speak a foreign language?

	D	<u>D</u>	TOTAL
F	50	10	60
<u>F</u>	20	20	40
TOTAL	70	30	100

Pay attention to
the keywords



Remember that earlier we said that Statistics is a linguistic problem. Here we found the first keyword that indicates that we have to use an axiom of probability.

Colloquial Language	Symbol	Probability	Operation
or	U	Union	Addition

We will use the
axiom of the sum

1. Probabilities

Contingency Tables: We use them when there is an idea of simultaneity in our data.

c) What is the probability of choosing a person who has a third level degree **or** speak a foreign language?

	D	<u>D</u>	TOTAL
F	50	10	60
<u>F</u>	20	20	40
TOTAL	70	30	100

Pay attention to
the keywords



We remember the Axiom of the sum: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Now we translate this to our exercise

$$P(D \cup F) = P(D) + P(F) - P(D \cap F) \rightarrow P(D \cup F) = 70/100 + 60/100 - 50/100 = 0.80$$

Answer: The probability of choosing a person who has a third level degree or speak a foreign language is 0.80

1. Probabilities

d) If we know that the person speaks a foreign language, what is the probability that also holds a third level degree?

Remember that earlier we said that Statistics is a linguistic problem. Here we found another keywords that indicates that we have to use an axiom of probability.

When we know something, we limit our total to that space instead of taking the universe.

The conditional axiom will help us here.

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Pay attention to
the keywords



1. Probabilities

d) If we know that the person speaks a foreign language, what is the probability that also holds a third level degree?

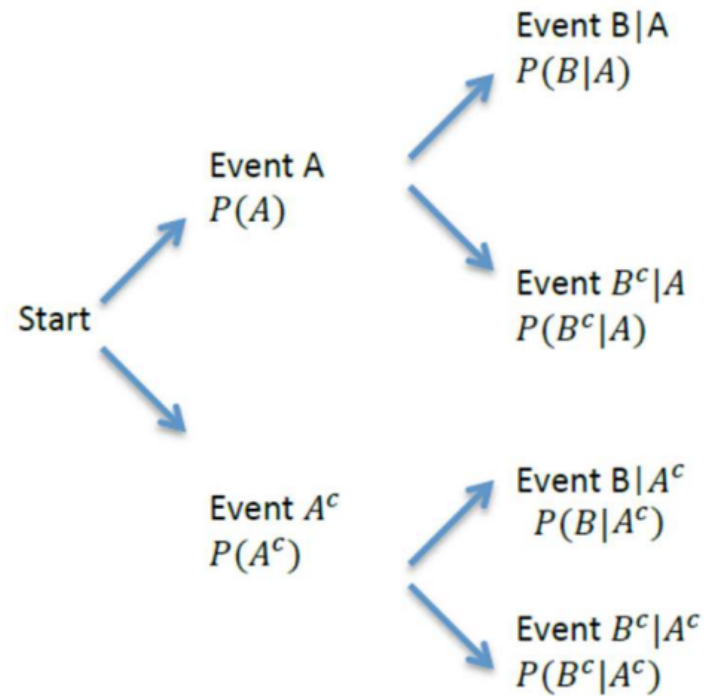
	D	<u>D</u>	TOTAL
F	50	10	60
<u>F</u>	20	20	40
TOTAL	70	30	100

We limit our analysis to this row, because we know that the person speaks a foreign language

$$P(D/F) = \frac{P(D \cap F)}{P(F)} = \frac{50}{60} = 0.8333$$

Answer: The probability of choosing a person that holds a third level degree, given that the person speaks a foreign language is 0.8333

1. Probabilities

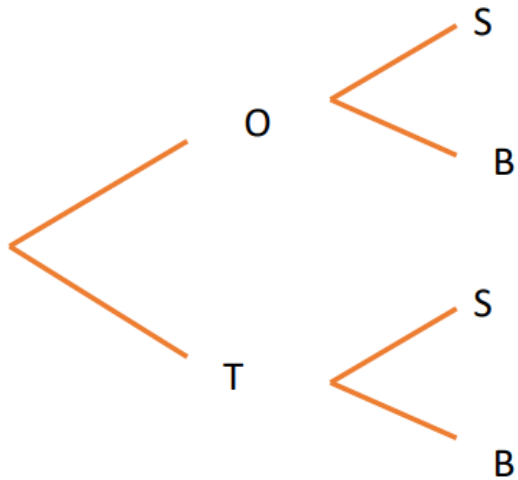


Tree Diagrams

1. Probabilities

Trees: We use them when there is an idea of **consecutiveness** in our data.

You find two purses. In one of them there are 5 silver coins and 3 bronze coins, and in the second one there are 4 silver coins and 6 bronze coins. If you picked a purse and took a coin from this one, what is the probability of picking a silver coin?



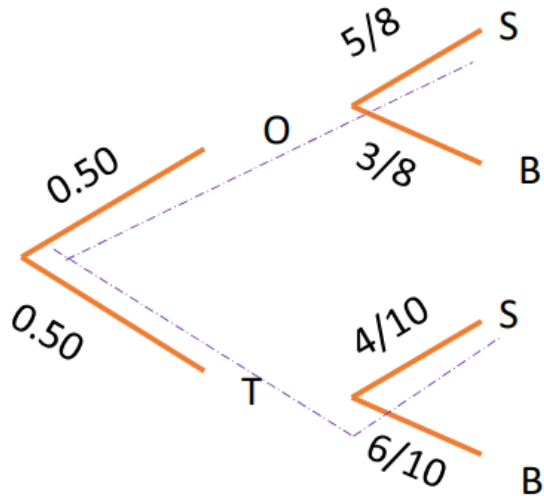
Once we have the tree, we need to add the probabilities on it



1. Probabilities

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We pick the purse One **AND** the coin is a Silver coin

OR

We pick the purse Two **AND** the coin is a Silver coin

$$P(S) = P(O \cap S) + P(T \cap S)$$

$$P(S) = 0.50 * 5/8 + 0.50 * 4/10 = 0.5125$$

↑
AND

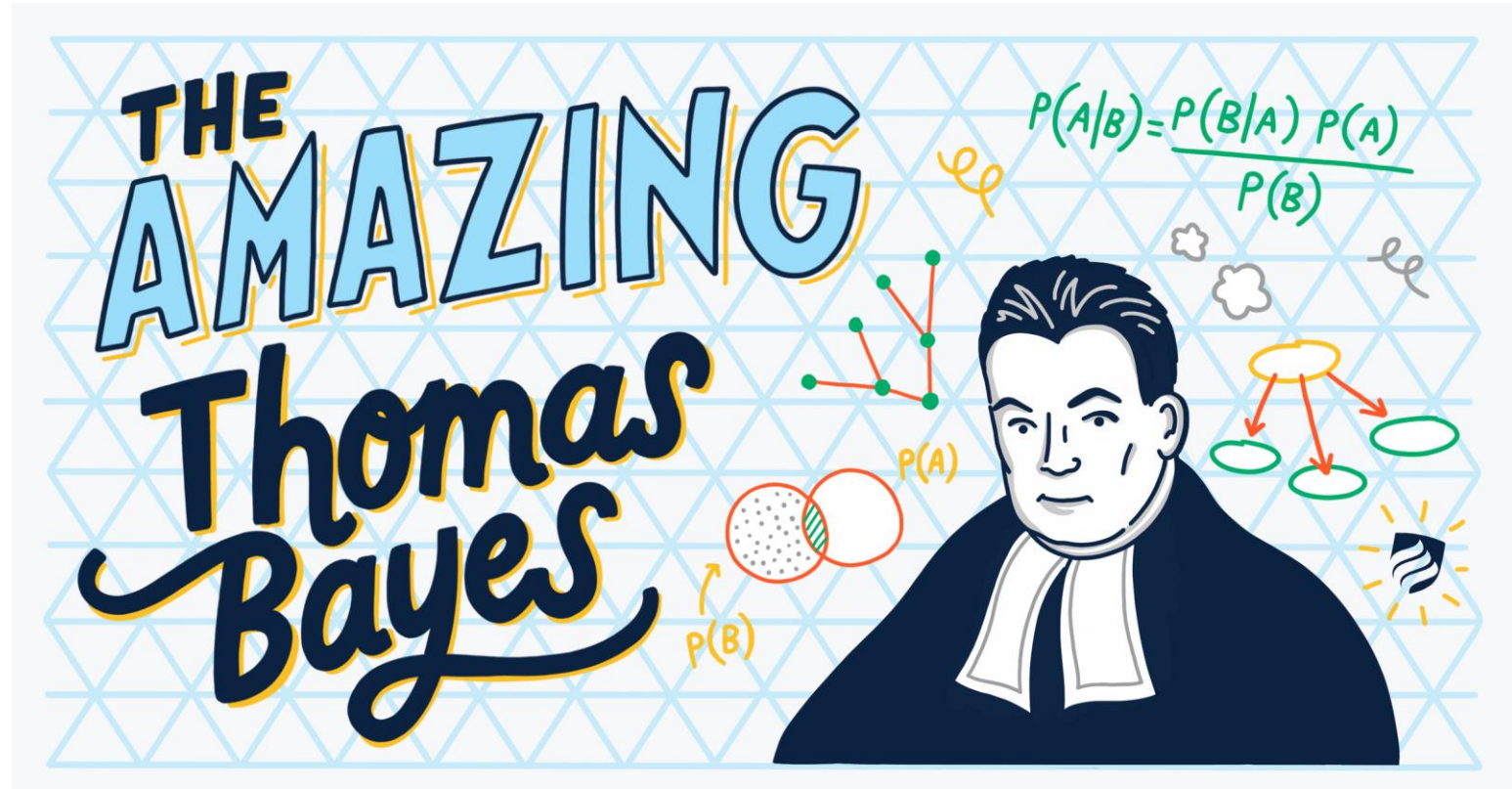
↑
OR

↑
AND

Answer: The probability of picking a silver coin is 0.5125

2. Bayes Theorem

Developed by Thomas Bayes, an eighteenth-century clergyman. One of the primary uses of Bayes's rule is to revise probabilities in accordance with newly acquired information. Such revised probabilities are actually conditional probabilities.



2. Bayes Theorem

Bayes's Rule

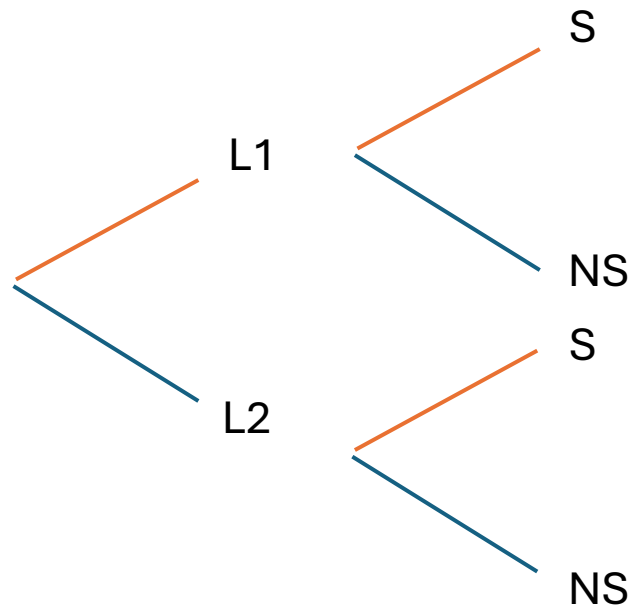
Suppose that events $A_1, A_2, \dots, A_k$ are mutually exclusive and exhaustive. Then for any event B ,

$$P(A_i | B) = \frac{P(A_i) \cdot P(B | A_i)}{\sum_{j=1}^k P(A_j) \cdot P(B | A_j)},$$

where A_i can be any one of events $A_1, A_2, \dots, A_k$.

2. Bayes Theorem

Smoking and Lung Disease According to the Arizona Chapter of the American Lung Association, 7.0% of the population has lung disease. Of those having lung disease, 90.0% are smokers; of those not having lung disease, 25.3% are smokers. Determine the probability that a randomly selected smoker has lung disease.



and

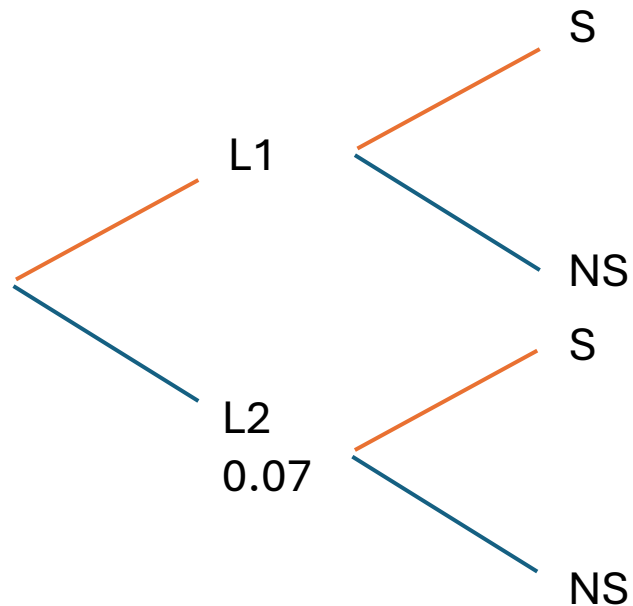
S = event the person selected is a smoker,

L_1 = event the person selected has no lung disease, and

L_2 = event the person selected has lung disease.

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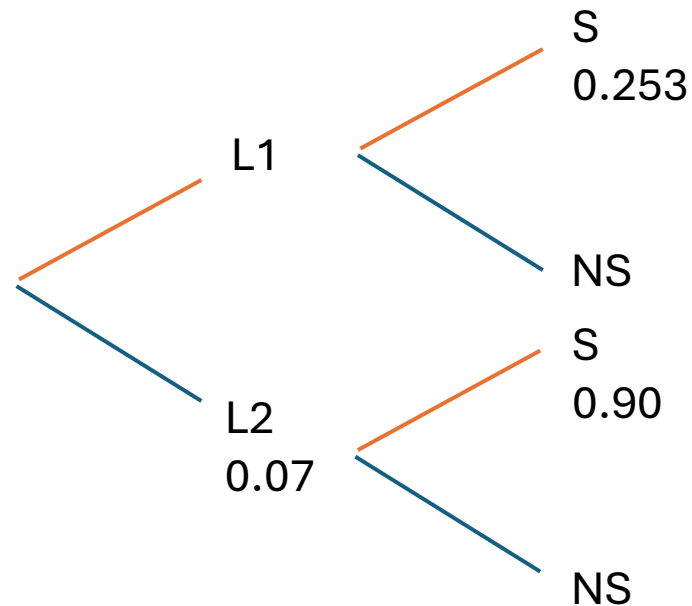
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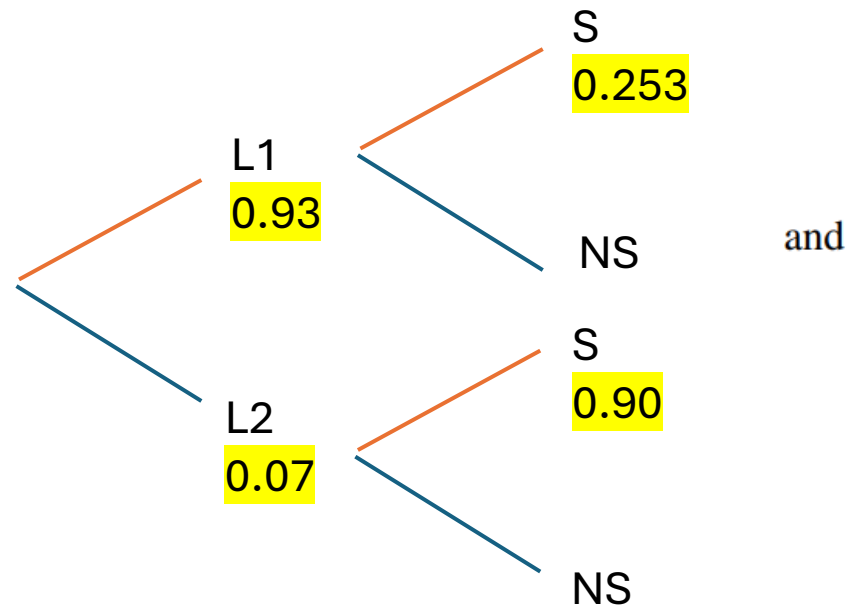
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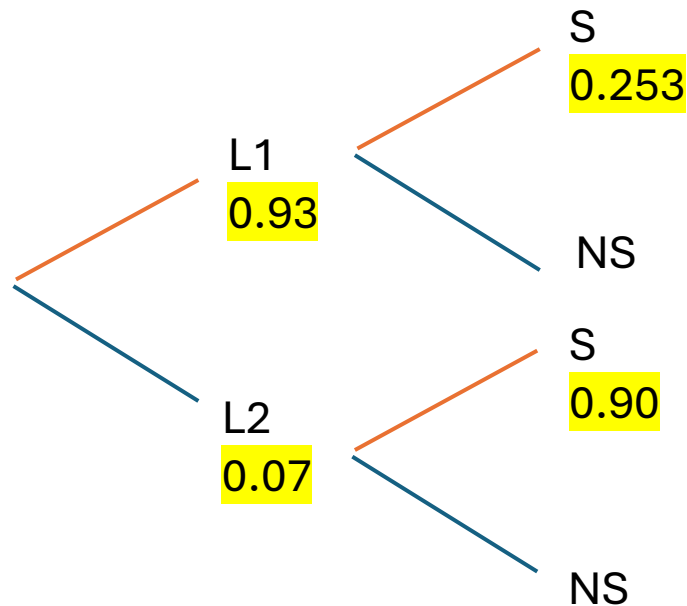
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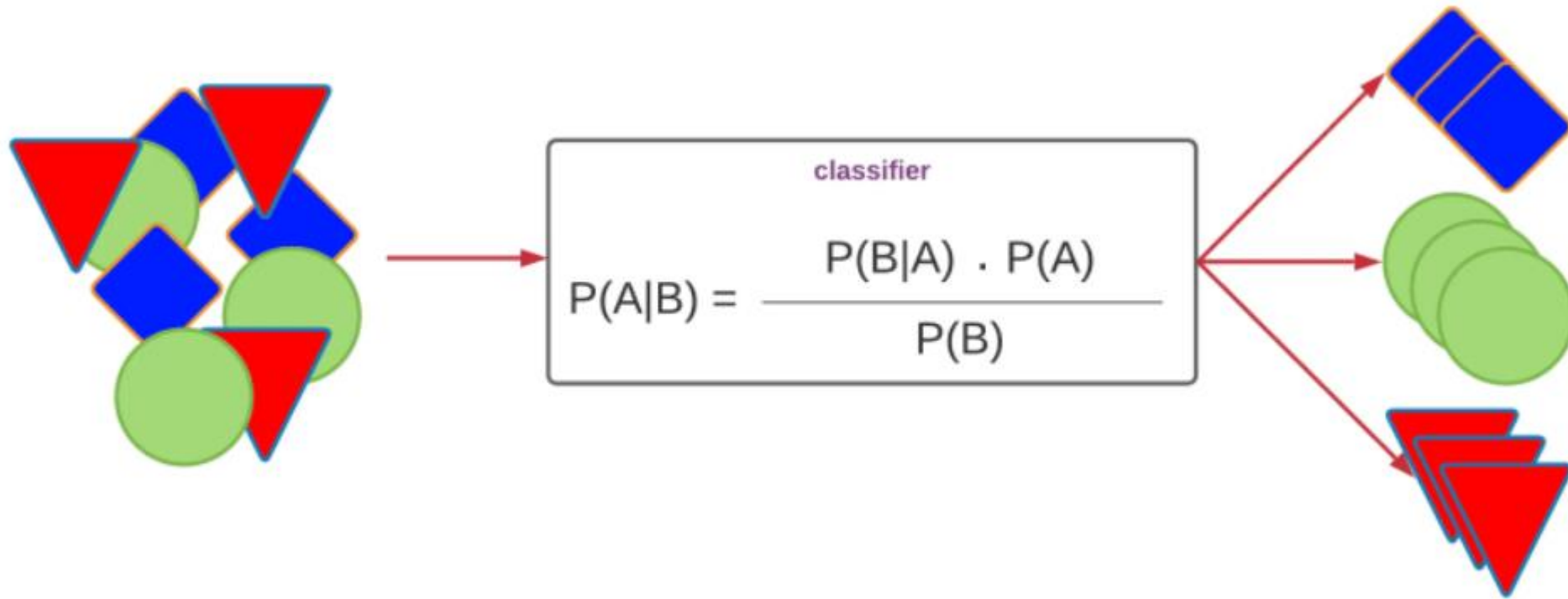
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$$\begin{aligned} P(L_2 | S) &= \frac{P(L_2) \cdot P(S | L_2)}{P(L_1) \cdot P(S | L_1) + P(L_2) \cdot P(S | L_2)} \\ &= \frac{0.070 \cdot 0.900}{0.930 \cdot 0.253 + 0.070 \cdot 0.900} = 0.211. \end{aligned}$$

3. ML Application



3. ML Application

```
# load the iris dataset
from sklearn.datasets import load_iris
iris = load_iris()

# store the feature matrix (X) and response vector (y)
X = iris.data
y = iris.target

# splitting X and y into training and testing sets
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.4, random_state=1)

# training the model on training set
from sklearn.naive_bayes import GaussianNB
gnb = GaussianNB()
gnb.fit(X_train, y_train)

# making predictions on the testing set
y_pred = gnb.predict(X_test)

# comparing actual response values (y_test) with predicted response values (y_pred)
from sklearn import metrics
print("Gaussian Naive Bayes model accuracy(in %):", metrics.accuracy_score(y_test, y_pred)*100)
```